

Md. Shalha Mucha Bhuyan

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 Shalha-Mucha18  Dhaka, Bangladesh

Research Interest

My research interests lie at the intersection of natural language processing and computer vision, with a focus on AI agents, Cybersecurity, and large language models to make them more robust, reliable, and safe.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh *2019 – 2024*
B.Sc in Electronics and Telecommunication Engineering (ETE)

- **GPA:** 3.20/4.0
- **Last Two Semesters GPA:** 3.40, 3.34

Undergraduate Thesis

Attention-Based Bengali Visual Question Answering: Enhancing Bengali Language Understanding of Multimodal AI Systems *June 2023 - June 2024*

- Developed a Bengali Visual Question Answering system that takes an image and a question in Bengali as input and produces contextually accurate answers by combining computer vision and natural language processing.
- Constructed a Bengali Visual Question-Answering (BVQA) dataset using zero-shot capabilities of LLMs to support research in low-resource language VQA.
- Proposed Multimodal Cross-Attention Network (MCRAN) leveraging pre-trained transformers and multi-head attention to fuse visual and textual features, outperforming baseline VQA methods.

Research Publications

- **BVQA: Connecting Language and Vision Through Attention for Open-Ended Question Answering** IEEE Access, Q1 journal, Impact Factor 3.4
Md. Shalha Mucha Bhuyan, Eftekhair Hossain, et al.
[DOI: 10.1109/ACCESS.2025.3540388](https://doi.org/10.1109/ACCESS.2025.3540388)  May 2025
- **Advancing Bangla Sign Language Recognition Through Deep Learning Techniques**
Fahmida Ahmed Ifty, **Md. Shalha Mucha Bhuyan**, et al.
[DOI: 10.1109/NCIM65934.2025.11159861](https://doi.org/10.1109/NCIM65934.2025.11159861)  IEEE NCIM, July 2025
- **Dataset for DDoS Detection in 5G-Enabled Connected and Autonomous Vehicle Systems**
Mirza Akhi, **Md. Shalha Mucha Bhuyan**, et al.
[DOI: 10.1109/GCWOT69191.2026.11499550](https://doi.org/10.1109/GCWOT69191.2026.11499550)  IEEE GCWOT, May 2026
- **GAN-Driven Defense for Securing Autonomous Vehicles from Emerging Cyber Threats**
Mirza Akhi, **Md. Shalha Mucha Bhuyan** et al.
[DOI: 10.1109/GCWOT69191.2026.11499711](https://doi.org/10.1109/GCWOT69191.2026.11499711)  IEEE GCWOT, May 2026

Experience

Software Engineer(AI/ML) *November, 2025 – Present*
Akij Resource, Dhaka 

- Conducted applied research in NLP, LLMs, Agentic AI, Multimodal AI, and Computer Vision, with a focus on model development, optimization, and real-world applications.
- Led end-to-end development of an agentic AI analytics platform(AR Next Intelligence) for the FMCG sector, designing multi-agent LLM workflows for automated KPI generation across 32 SBUs, reducing reporting latency by 30% and improving forecasting accuracy by 40%.
- Implemented a multi-agent workflow using LangGraph, integrating tools and agents to deliver automated business insights to end users.
- Deployed LLM and AI application workloads on production-grade AWS infrastructure (ECR, EKS), enabling automated deployment and scalable service delivery.
- Built a production-grade RAG pipeline for AI agents with vector databases, reducing retrieval times and improving overall LLM response quality.

Software Engineer Intern (AI/ML)

March – October, 2025

Cloudly Infotech Limited, Dhaka Bangladesh [🔗](#)

- Developed **Maveric Copilot**, a telecommunications-focused agentic orchestration system for real-time network log analysis and automated report generation.
- Developed agentic AI workflows and orchestration pipelines using frameworks such as LangChain, CrewAI, and LLM-based task automation, enabling dynamic multi-agent collaboration and seamless tool integration.
- Fine-tuned LLMs for domain-specific business tasks to improve task performance and relevance.

Technical Skills

Programming Languages:	Python, C, C++, JavaScript, SQL
Data Analysis:	Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, SQL
Database Management:	PostgreSQL, Vector Databases (Pinecone, FAISS)
AI/ML:	TensorFlow, PyTorch, Scikit-Learn, LangChain, LangGraph, MCP Server
Backend:	FastAPI, Django, jwt, OAuth
DevOps:	Docker, Kubernetes, Terraform, Bash Scripting, Ansible, Prometheus
MLOps:	MLflow, DVC, DagsHub
Cloud Technologies:	AWS, GCP
Tools & Platforms:	VS Code, PyCharm, GitHub, Latex

Certifications

AWS Cloud Technical Essentials (AWS), **Agentic AI** (DeepLearning.ai), **Fundamentals of MCP** (Hugging Face), **Generative AI with Large Language Models** (Coursera), **TensorFlow Developer Professional Certificate** (Coursera)

Achievements

- **10th place in ML Competition**, CUET-ETE Day

Extracurricular activity

- Co-Finance Secretary, CUET Sports Club
- Contest Management Secretary, CUET ETE Day
- Champion, ETE Premier League (EPL), organized by the Department of ETE
- Attended Cultural Programme at ETE Day 2022, organized by the Department of ETE

Reference

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